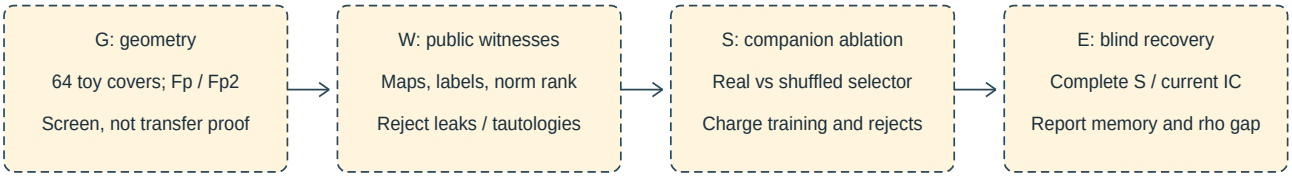
Prym experiment programme: issue #2414

Prospective designs and software components. No scientific trial or measured gain.



Proposed tests. No measured gain. Admission and independent review remain required.

Designed for issue #2414 and PR #2415. **Prospective designs; no scientific trials or measured advantage.** The four experiment contracts are review_required; native admission is required before execution. This extends the intake under existing proposal IDEA-20260909-4c5ee9.

The diagram shows the decision sequence; report.pdf is derived by render_report.py. Full scientific contracts live in the linked specifications, not this communication report.

G — Verify the geometry and screen direct transfer

EXP-ECDLP-da4ba3, H-ECDLP-9e7a19.

For $E:y^2=x^3+7$, use eight deterministic quadratic $h=(x-a)(x-b)$ at each of eight primes 11 through 59. Require distinct roots and $\gcd(f,h)=1$. These are **64 newly specified toy covers**, unrelated to the unidentified “32 covers.” D is the smooth fiber product $y^2=f$, $z^2=h$; $H:w^2=fh$ models its genus-two complement up to isogeny.

Count E, H and D over Fp and Fp^2 by enumerating square multiplicities. Check N_D=N_E+N_H-(p^m+1) for m=1,2; reconstruct F_E and F_A, factor F_A, and compute gcd over Q. Also report whether the largest odd prime divisor r of #E divides #A. These are different screens: Frobenius detects possible homomorphisms; group order detects possible rational r-torsion.

Synthetic coprime/shared polynomial cases test the gcd instrument only. Ordinary and supersingular source cases are labelled separately. Two trace checks do not independently verify the complete genus-three polynomial. No Jacobian map or efficient transfer evaluator is certified by this pilot. Passing enables map/interface validation; it supplies no attack evidence.

Implemented: geometry_pilot.py and frozen prospective geometry_inputs.json. Only compilation and CLI checks were performed this turn. Runtime wrapper, archive and admission remain due.

W — Audit source witnesses, independent columns, and native rank

EXP-ECDLP-ad0ffe, H-ECDLP-51f052.

A known-source constructor receives a public a and R=[a]P, frozen factor-base atoms and cover data. It must return a checked divisor relation whose norm has a known coefficient of P. A blind constructor receives Q+[t]P with t public and the same base. Neither receives the secret k or factor-base logs.

Begin with the explicit PO-transfer-003 principal-divisor generator as an instrument calibration. Use its two maps on $C:y^2=x6+A*x^2+B$: $pi1=(x^2,y)$ to $E1:Y^2=U^3+A*U+B$, and $pi2=(B/x^2,B*y/x^3)$ to $E2:V^2=W^3+A*W^2+B^2$. Here the Prym is elliptic; this is a different cover family from G, and its maps need their own verification.

Check every source row independently on E1. Normalize duplicate norm images, signs and zero columns before rank modulo r. Derive the required rank from the actual unknown columns and gauge constraints; do not assume “B-1.” Reject wrong signs, false labels, post-hoc kernel residuals, and tuples whose source log is unknown. Planted known-log atoms test correctness only and are excluded from decisive unknown-log rank and recovery panels.

Positive: an admissible generator supplies independently useful public rows. Negative: identify the exact leakage, tautology, map error or rank defect. Neither outcome closes all Prym approaches. The compact new router remains open; this audit does not reopen historical IDEA-002.

S — Test one explicit auxiliary-component selector

EXP-ECDLP-5e7e74, H-ECDLP-7cd883.

For each PO003 cubic-interpolation candidate, sum the four interpolation points after mapping them to E2. Before solving the residual quadratic, bucket this auxiliary sum by infinity, or by (Legendre(W), floor(16*W/p)). This is a testable public feature; no geometric argument currently says it predicts factor-base acceptance. Computing it may cost more than the work it avoids.

Use disjoint hash partitions of complete candidate identities. Train on 2^16 distinct tuples and evaluate on 2^16 fresh tuples per admitted instance. Canonicalize effective source coefficients, FB permutations and the known acceptance-preserving involutions before hashing, so geometric aliases stay in the same partition. This integration is still required. Rank nonempty bins by (hits+1)/(attempts+2), retain their top quarter rounded up, break ties by canonical bin string and reject unseen bins. A hit requires both residual endpoints in the exact FB and a valid public source row. Freeze the model before holdout. Insufficient support is inconclusive.

Compare identical candidate streams under no filter, identity-selector A/A, the real filter, and a label-permuted filter. Separately test a split-product null and transported coordinate data; these diagnose false explanations. Independent curves are the inferential units. Acceptance precision and recall are diagnostic; decisive metrics are charged work per independent useful row and, eventually, complete S at the same completion target.

The selector component implements the maps, group arithmetic, bins, exact model ranking and record-level split checks. Its two software tests passed. Generator integration and the current IC baseline are still missing. This tests an engineering improvement within a fixed generator's scope; it does not escape P1546 or establish an improved asymptotic exponent.

E — Attempt blind recovery with complete accounting

EXP-ECDLP-e193fa, H-ECDLP-ad8992.

Run only after source/rank validity and a genuine paired cost benefit survive independent review. Proposed sizes are 11,13,15,17 bits of subgroup order r, ten fresh instances per size and three paired seeds: 120 paired attempts per arm. Freeze actual curve constants, generators, targets, search failures, source/solver hashes and baseline receipts before approval. The challenge process alone holds k; it reveals k only after sealed attack output.

Charge construction/search, training, preprocessing, relation attempts and rejects, normalization, rank, linear algebra, blind descent, recovery and final verification. Report peak memory, single-target setup-inclusive S, and any separately specified amortized scope. A subroutine gain is not an end-to-end gain.

Success requires an independently verified complete-S improvement over the CURRENT matched IC baseline, beyond A/A variation, with correct blind recovery and the prescribed memory allowance. Report the distance to matched rho separately. Rho is not the incremental pass/fail threshold. Failures and censoring remain in the denominator; do not average successful runs alone. Toy scaling does not certify security impact on secp256k1 or P-256.

Admission and next work

The exact native diagnostic reports and director assessment are linked from admission.json. Missing integrations and immutable runtime bindings leave these protocols review_required, never silently approved. No runbook is committed unless it can lint cleanly from complete inputs. Next implement the PO003 candidate/row checker and metering integration, bind the current verified IC pipeline, and materialize the prospective fixture sets. The original historical contract and existing scientific states are preserved.

dominated_by: not assessed; no algorithmic result claimed. sota_delta: no measured gain; four designs and prospective software components. Canonical scientific graph nodes were not promoted: this graph records proposed test dependencies, not established reduction edges.